Wholesale Distribution Analytics: 2-Year Performance Review and Strategic Insights

£17.4M TOTAL REVENUE (2 YEARS)	**36,969** TOTAL ORDERS	**5,878** UNIQUE CUSTOMERS
£470 AVERAGE ORDER VALUE		

Key performance indicators across 41 countries, December 2009 to December 2011.

Executive Summary

This analysis examines transactional data from a multi-country B2B wholesale retailer spanning December 2009 to December 2011. The project identifies revenue trends, customer behavior patterns, product performance, and operational opportunities that support strategic decision-making.

Problem: The organization faced fragmented visibility into customer value, product profitability, seasonal demand, and geographic performance—limiting ability to optimize inventory, prioritize customer retention, and forecast revenue.

Solution: We conducted comprehensive data cleaning (removing 27% of raw records due to cancellations, missing customer IDs, and duplicates), exploratory analysis, and customer segmentation using RFM (Recency, Frequency, Monetary) methodology.

Key Findings: - **Extreme customer concentration:** Top 20% of customers drive 77% of revenue, creating significant retention risk and opportunity. - **Strong seasonal pattern:** November emerges as the peak revenue month in both 2010 and 2011 (>£1.15M each), with Q4 showing consistent strength. - **Dominated by low-price products:** Items priced below £5 account for 79% of revenue; premium pricing (>£10) contributes only 8%. - **Dormant customer base:** Approximately 51% of customers showed no purchase activity in the past 90 days, indicating engagement risk. - **Geographic concentration:** UK accounts for 83% of revenue; Ireland, Netherlands, and Germany are the leading non-UK markets.

Impact: Implementing targeted customer retention programs, inventory planning tied to seasonal demand, and focused international market development can protect revenue, optimize operations, and unlock growth in underpenetrated markets.

Data Overview and Cleaning

Dataset: Online Retail II (UCI Machine Learning Repository) containing 1,067,371 raw transaction records from a UK-based online retail operation.

Period: December 2009 to December 2011 (25 months)

Scope: 41 countries, 5,315 unique products, 5,878 customers (after cleaning)

Data Quality Issues Addressed:

Issue	Records Removed	Resolution
Cancelled invoices (prefix 'C')	19,494	Excluded as returns, not revenue events
Missing CustomerID	242,257	Excluded from customer-level analysis
Non-positive unit prices	71	Removed as data errors
Duplicate records	26,124	Removed as system artifacts
Total rows removed	**287,946 (27%)**	—

Final Dataset: 779,425 clean transactions representing £17.4M in revenue across 36,969 distinct invoices.

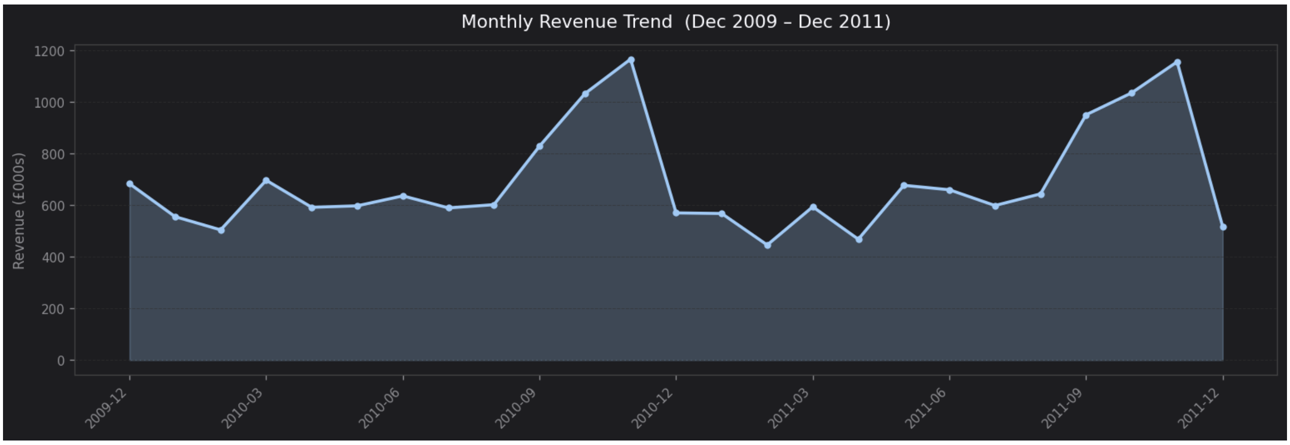
Data Dictionary: All analysis uses standardized fields including InvoiceNo, StockCode, Description, Quantity, UnitPrice, CustomerID, Country, and InvoiceDate. Derived fields include Year, Month, YearMonth, and Revenue (Quantity × UnitPrice). No currency conversion applied; all values in GBP.

Sales Performance and Seasonal Analysis

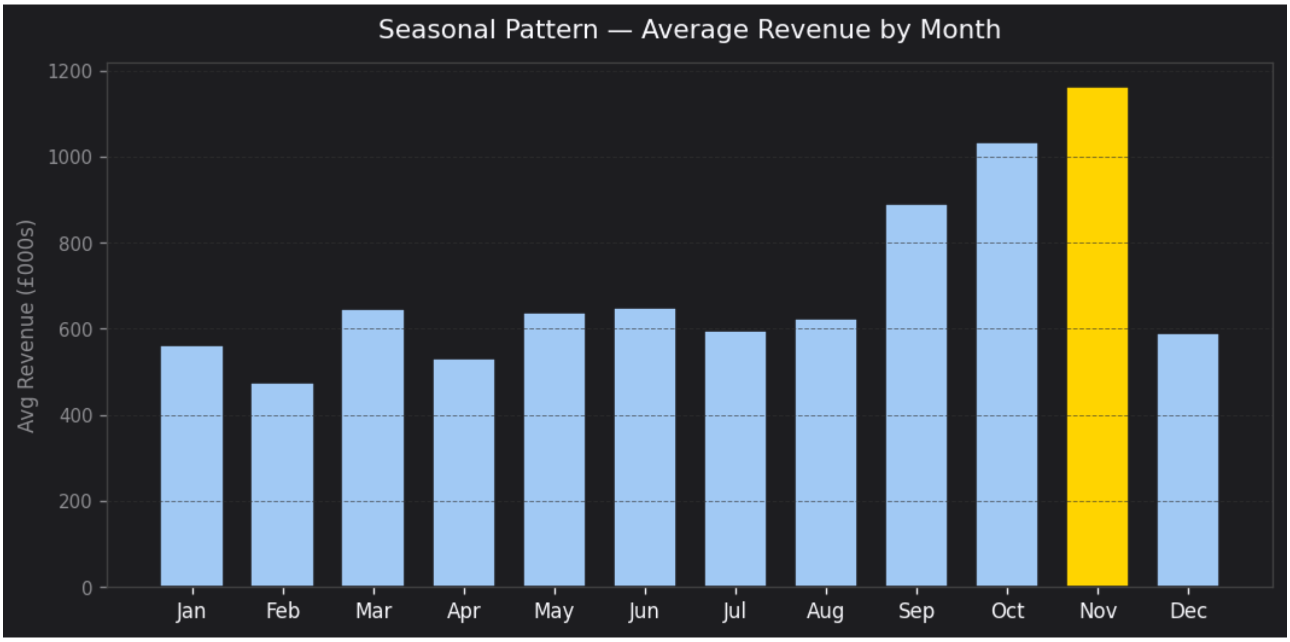
Revenue analysis across 25 months reveals clear seasonal patterns tied to Q4 demand peaks. November consistently outperforms all other months, with peak revenue in November 2010 (£1.17M) followed by November 2011 (£1.16M). October also shows elevated activity (£1.03M average), suggesting early holiday purchasing or year-end business cycles.

Conversely, February records the weakest performance (£475k average), with April also showing depressed activity (£530k). This 2.4x variance between peak and trough months indicates strong seasonality that should drive inventory planning and staffing decisions.

The top 5 revenue months are dominated by Q4 activity, accounting for £5.35M out of £17.4M total revenue (31% concentration in just 5 months). This concentration creates both planning opportunities and cash flow management needs.



Monthly revenue trend spanning December 2009 to December 2011. Peak activity occurs in November of both years (£1.17M and £1.16M), with lowest revenue in February. The pattern demonstrates consistent year-over-year seasonality and recurring peaks approximately 13 months apart.



Average monthly revenue pattern reveals November as the dominant revenue driver (£1.16M), followed by October (£1.03M). A secondary peak appears in September (£900k). The low season spans February through April, with February the weakest month (£475k). This consistent pattern across both years supports forecasting and resource planning decisions.

Customer Analysis and Segmentation

RFM (Recency, Frequency, Monetary) segmentation identified 8 customer segments, with stark differences in revenue contribution and engagement risk. This analysis reveals critical customer concentration and dormancy concerns.

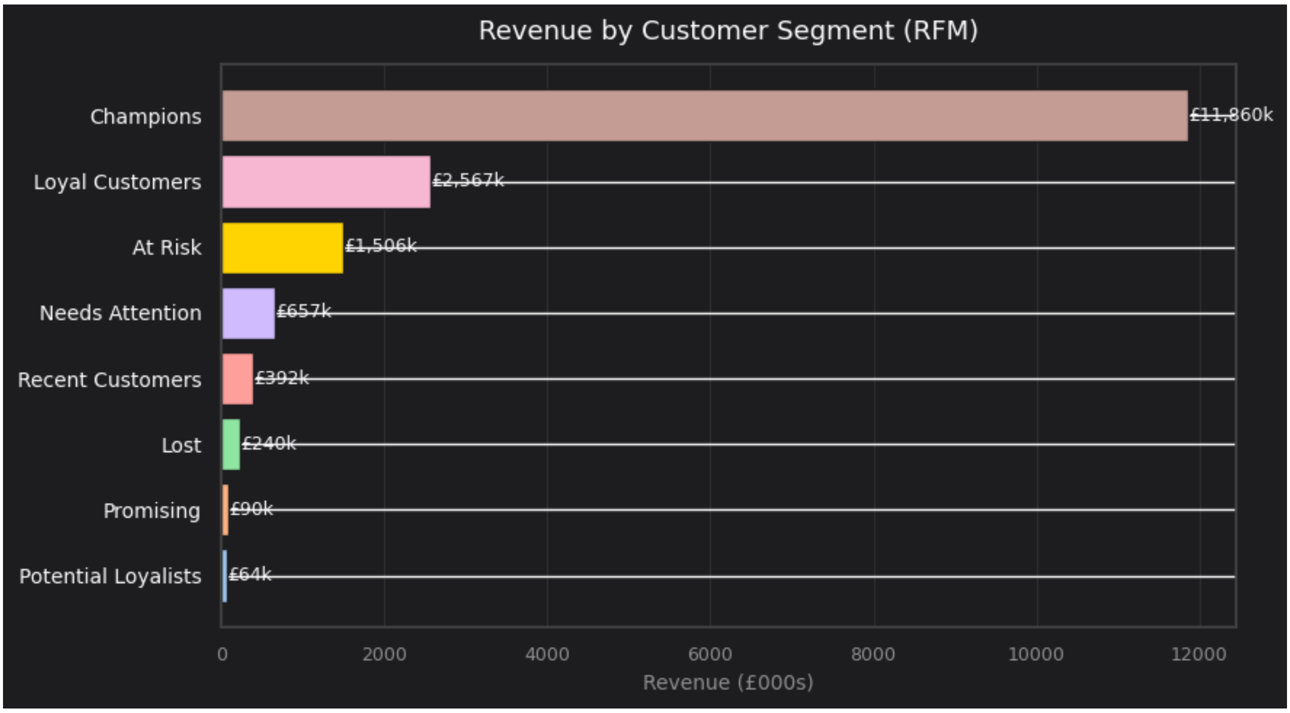
Segment Breakdown:

Segment	Count	Revenue	% of Total	Risk Level
Champions	1,297	£11.86M	72.4%	Low (high engagement)
Loyal Customers	1,138	£2.57M	15.7%	Low-Medium
At Risk	616	£1.51M	9.2%	**High** (declining engagement)
Needs Attention	1,716	£657k	4.0%	**Critical** (dormant)
Recent Customers	443	£392k	2.4%	Medium (new, unproven)
Lost	465	£240k	1.5%	**Critical** (churned)
Promising	173	£99k	0.6%	Medium
Potential Loyalists	30	£64k	0.4%	Low-Medium

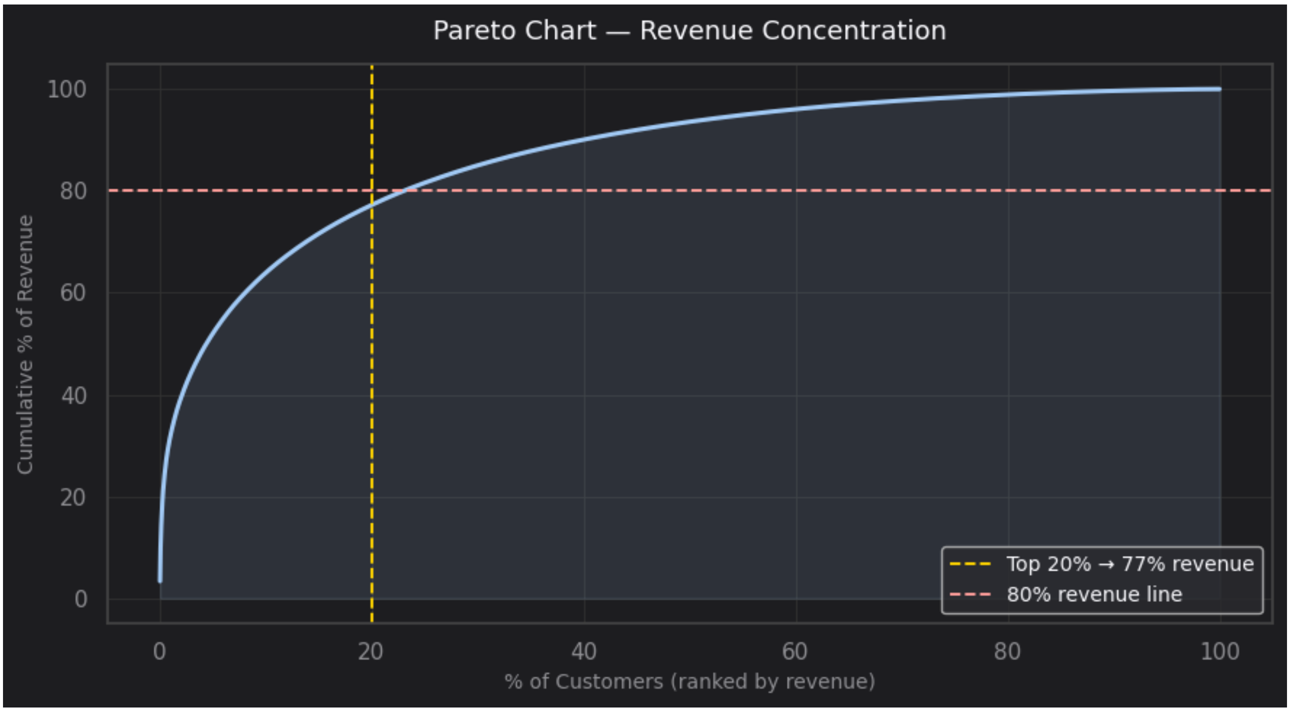
Key Insights:

- **Extreme concentration:** The top 20% of customers (Champions + Loyal Customers) generate 88.1% of revenue. Losing just a handful of Champions poses existential revenue risk.
- **Large dormant base:** 50.8% of all customers (2,985 individuals) have not purchased in over 90 days, indicating significant re-engagement opportunity or acquisition decay.
- **Top 10 customers alone:** The highest-value customer (ID 18102) has generated £581k in lifetime revenue with 145 transactions. The top 10 customers represent £3.2M in revenue (18% of total), underscoring concentration.

Business Implication: Customer retention programs must prioritize Champions and Loyal segments. Simultaneous re-activation campaigns for the At Risk and Needs Attention segments (2,332 customers, £2.2M at stake) could materially improve profitability.



Customer segment revenue contribution. Champions dominate with £11.86M (68% of total), while Loyal Customers contribute £2.57M (15%). Combined, these two segments represent 83% of all revenue. Smaller segments (At Risk, Needs Attention, etc.) collectively represent <£2.5M despite representing 70% of customer count, illustrating extreme revenue concentration.



Pareto curve demonstrating extreme revenue concentration. The top 20% of customers generate 77% of total revenue, confirming the classic 80/20 principle. This visualization highlights the critical importance of retention: loss of just 5% of the customer base risks 40%+ of revenue. Conversely, it shows significant opportunity to develop mid-tier customer segments.

Product Performance Analysis

Product portfolio analysis spans 5,315 SKUs, revealing significant concentration in top performers and a long tail of slow-moving inventory.

Top Performers:

The top 20 products generate approximately £3.0M in revenue (17% of total), led by home décor items:

1. **Regency Cakestand 3 Tier** – £278k (3,317 orders, repeat rate 2.52x)
2. **White Hanging Heart T-Light Holder** – £247k (4,888 orders, repeat rate 3.28x)
3. **Paper Craft, Little Birdie** – £168k (1 order, likely bulk purchase)

High-Repeat Products:

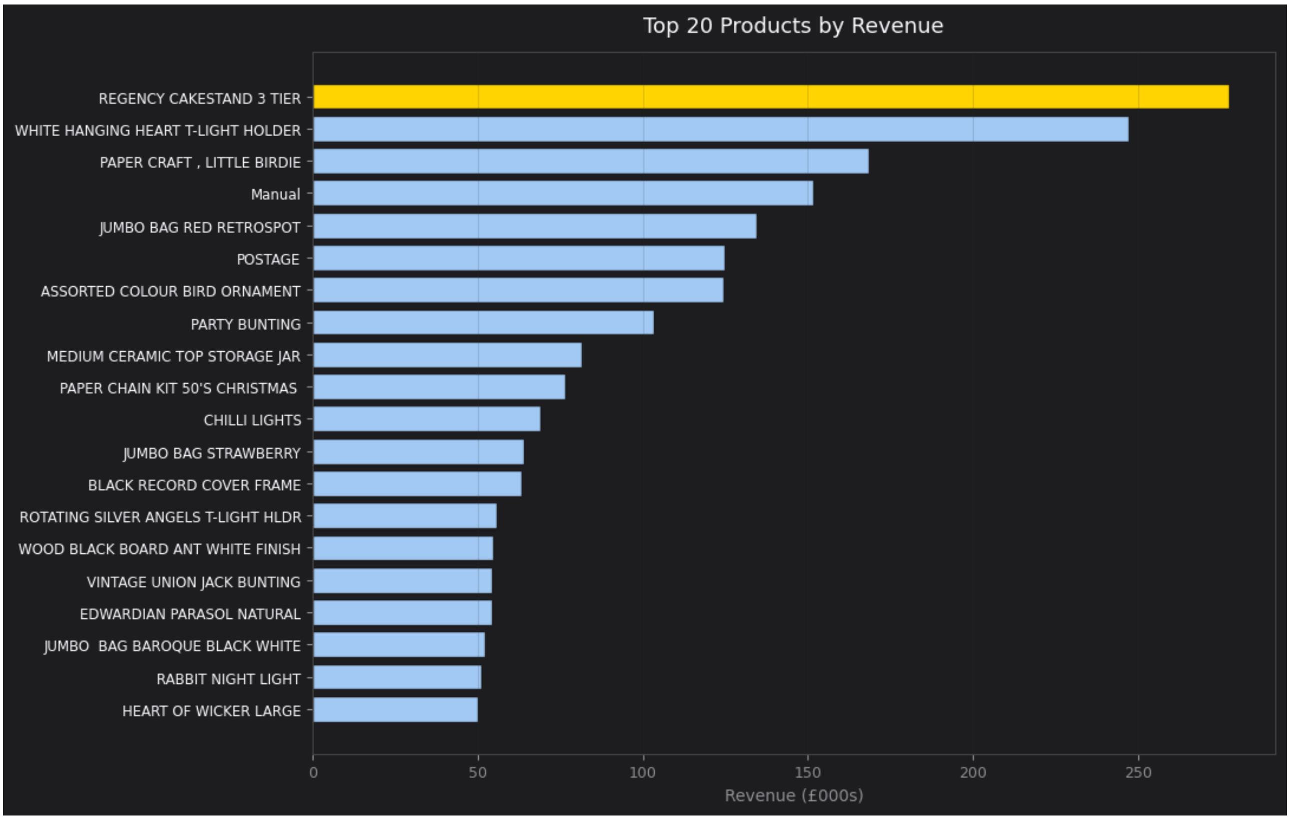
Products with strong repeat purchase rates indicate customer satisfaction and revenue stability: - Postage: 4.45x repeat rate, 1,803 orders, £125k revenue - White Hanging Heart T-Light Holder: 3.28x repeat rate, 4,888 orders - Jumbo Bag Red Retrospot: 3.04x repeat rate, 2,612 orders

These high-repeat items should receive priority in inventory planning and marketing.

Slow-Moving Inventory:

- **374 products** (7% of catalog) have been ordered ≤2 times in 25 months, representing dead weight in inventory management.
- These SKUs likely tie up working capital and warehouse space without meaningful revenue contribution.
- **Recommendation:** Conduct detailed profitability analysis on slow-movers to inform discontinuation or bundling strategies.

Product Diversity: Despite 5,315 SKUs in the catalog, the business operates efficiently through a concentrated portfolio of bestsellers, allowing focused customer acquisition and fulfillment optimization.



Top 20 products by revenue. Regency Cakestand 3 Tier dominates at £278k, followed by White Hanging Heart T-Light Holder (£247k) and Paper Craft Little Birdie (£168k). Revenue distribution among top 20 shows clear hierarchy with significant drop-off after position 3. Products 4–20 each generate £75k–£155k, indicating a broad base of solid performers beneath the top tier.

Pricing Analysis

Pricing strategy centers on volume-driven, low-price-point products. This approach drives transaction volume and market reach but constrains margin per unit.

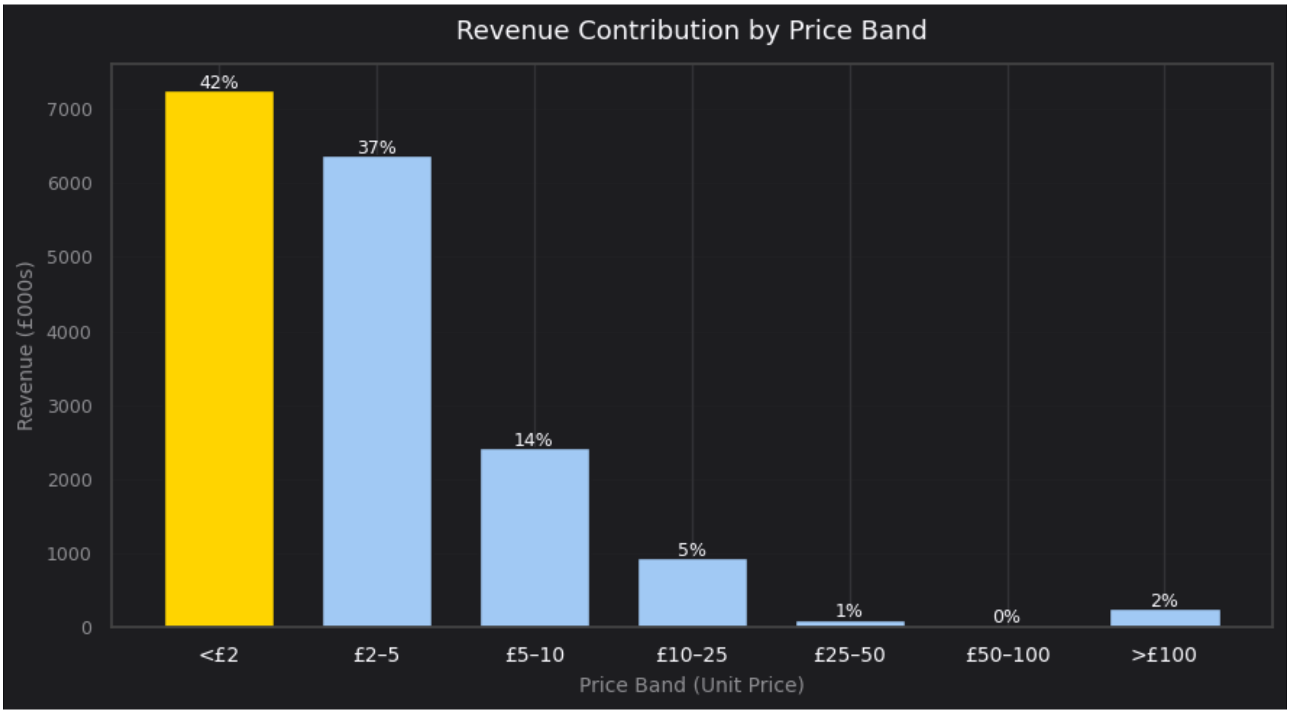
Revenue by Price Band:

Price Band	Revenue	% of Total	Orders	Avg Price
<£2	£7.25M	41.7%	32,881	£1.13
£2–5	£6.37M	36.7%	31,710	£3.24
£5–10	£2.42M	13.9%	23,638	£7.51
£10–25	£934k	5.4%	12,374	£13.84
£25–50	£107k	0.6%	1,085	£35.86
£50–100	£34k	0.2%	395	£59.89
>£100	£255k	1.5%	329	£775.61

Key Insights:

- **Extreme skew toward budget pricing:** Items priced below £5 generate £13.6M (78.5% of revenue) from 64,591 orders, demonstrating that volume-based pricing dominates the business model.
- **Low correlation between price and purchase frequency:** Price-purchase frequency correlation coefficient of -0.032 indicates virtually no relationship between item price and repurchase rate, suggesting customer buying decisions are driven by product fit rather than price sensitivity.
- **Premium segment opportunity:** The >£100 price band, while small in volume (329 orders), generates disproportionate revenue (£255k, 1.5%), highlighting margin potential if scalable to high-volume segments.
- **Mid-range underperformance:** The £10–100 bands generate only 8.2% of revenue despite representing 14,253 orders, suggesting potential to bundle or promote higher-margin items.

Recommendation: Given the low price-frequency correlation, consider premium product line expansion alongside core low-price offerings to improve average order value without sacrificing volume.



Revenue contribution by price band demonstrates the business's reliance on low-price-point items. The <£2 and £2–5 bands combine for 78.5% of revenue, while all premium bands (>£10) contribute only 8.2%. This distribution reflects a volume-driven business model optimized for market penetration over margin maximization.

Geographic Market Performance

The company operates across 41 countries with significant geographic concentration in the UK and selected European markets. While the UK dominates by volume, international markets show promising growth trajectories.

Market Breakdown:

Domestic (UK): £14.4M revenue (82.8%), 33,541 orders, 5,350 customers, £430 average order value

International (40 countries): £2.99M revenue (17.2%), 3,428 orders, 528 customers, £872 average order value

Top 10 International Markets:

Rank	Country	Revenue	Orders	Customers	Avg Order Value
1	Ireland	£617k	567	5	£1,088
2	Netherlands	£554k	228	22	£2,422
3	Germany	£425k	789	107	£539
4	France	£349k	614	95	£569
5	Australia	£169k	95	15	£1,784
6	Spain	£108k	154	41	£702
7	Switzerland	£100k	90	22	£1,111
8	Sweden	£92k	104	19	£882
9	Denmark	£69k	43	12	£1,604
10	Belgium	£65k	149	29	£437

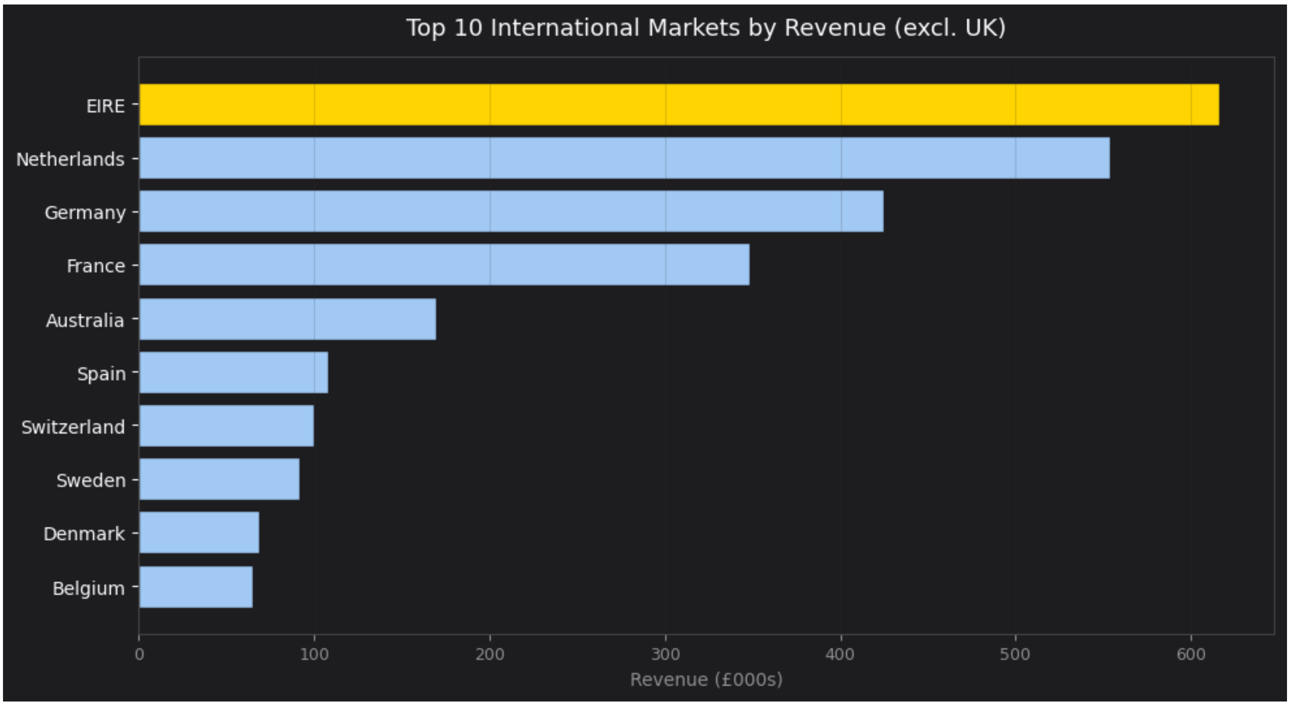
Growth Markets (2010→2011 YoY):

Fastest-growing markets among those with >£10k 2010 base revenue: - **Australia:** +336% growth (£31.5k → £137.5k) - **Japan:** +123% growth (£13.3k → £29.7k) - **Belgium:** +54% growth - **Spain:** +47% growth - **Portugal:** +42% growth - **France:** +39% growth

Strategic Insights:

- **Untapped opportunity:** Australia's 336% growth suggests successful market entry potential. The low customer base (15) against high revenue (£169k) indicates high-value accounts rather than broad penetration.
- **European stability:** UK and core European markets (Germany, France, Netherlands) show consistent, stable revenue with less volatility, providing revenue predictability.
- **Emerging potential:** Asian markets (Japan +123% growth) warrant continued investment despite low absolute size.

Recommendation: Increase investment in fast-growing markets (Australia, Japan, Benelux) while maintaining UK customer base. Consider localized marketing campaigns for high-growth regions.



Top 10 non-UK markets by revenue. Ireland leads at £617k, followed by Netherlands (£554k) and Germany (£425k). European markets comprise 8 of top 10, with Australia (£169k) and Japan the exceptions. Geographic revenue concentration in Western Europe reflects established market presence and cultural/linguistic proximity to UK operations.

Operational Insights and Anomalies

Operational analysis reveals ordering patterns, bulk purchaser behavior, and anomalous transactions that warrant management attention.

Large Invoice Analysis:

Using statistical outlier detection (invoices exceeding 3× interquartile range): - **Outlier threshold:** £1,435 per invoice - **Unusual large invoices:** 1,439 invoices (3.89% of all orders) exceed this threshold - **Largest invoice:** £168,470 (invoice 581483, customer 16446, UK, 80,995 units of single item on 2011-12-09)

Top 3 largest invoices: 1. Customer 16446: £168,470 (80,995 units, single line item) – likely bulk distributor or reseller 2. Customer 12346: £77,184 (74,215 units) – bulk purchase pattern 3. Customer 14156: £44,052 (94 line items, mixed products) – wholesale distributor

Bulk Purchaser Segment:

Identified 10 customers with avg quantity >45 units per line item (top 5% threshold): - Customer 13902: 3,501 units per line (5 orders, 220,600 total units) - Customer 13687: 1,937 units per line (1 order, 87,167 total) - Customer 17940: 2,433 units per line (16 orders, 85,148 total)

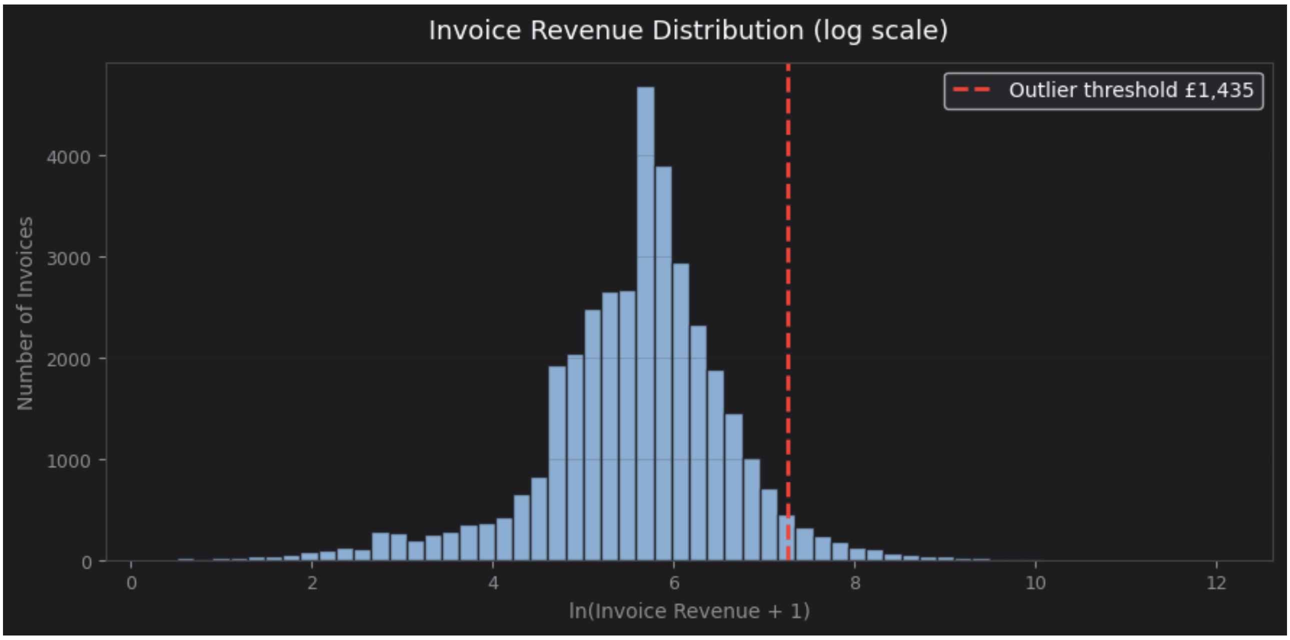
Peak Ordering Periods:

Daily order volume averages 61.2 orders (std dev 23.8). Three spike days recorded with >3σ deviation: - **2010-11-25:** 146 orders (2.38σ above mean) - **2010-12-02:** 137 orders (1.24σ above mean) - **2011-11-17:** 136 orders (1.22σ above mean)

These spike days correlate with mid-November/early December holiday purchasing cycles.

Operational Implications:

- **Bulk purchaser risk/opportunity:** 10 bulk purchasers generate significant revenue but create dependency. Account management should assess sustainability.
- **Seasonal planning:** November 25 and December 2 peaks demonstrate need for contingency staffing and fulfillment capacity during holiday season.
- **Inventory positioning:** The concentration of spike days in Q4 reinforces need for pre-season inventory builds and working capital planning.



Invoice revenue distribution plotted on logarithmic scale reveals a log-normal distribution with most invoices clustered £400–£800. The red dashed line at ln(£1,435) marks the statistical outlier threshold. Approximately 1,439 invoices (3.89%) exceed this threshold, representing bulk orders, reseller purchases, or institutional accounts requiring special handling.

Strategic Recommendations

Based on the analysis, the following actions will drive measurable business impact:

1. Customer Retention Programs (High Priority)

Problem: Top 20% of customers drive 77% of revenue; 51% of customer base is dormant (>90 days).

Actions: - Establish dedicated account management team for Champions segment (1,297 customers generating £11.9M). Implement quarterly business reviews, personalized communication, and early warning system for churn indicators. - Launch re-engagement campaign for "At Risk" segment (616 customers, £1.51M revenue). Deploy targeted discounts, product recommendations, or account reviews. - Implement automated dormancy alerts for customers approaching 90-day threshold. Trigger personalized outreach from sales team.

Expected Impact: 10% reduction in churn among top 20% could protect £1.7M in annual revenue.

2. Seasonal Inventory Planning (Medium Priority)

Problem: November revenue is 2.4x February revenue; Q4 accounts for 31% of annual revenue concentrated in 5 months.

Actions: - Build inventory models forecasting Q4 demand (peak months: September–December). Increase safety stock by 20–30% by August. - Plan staffing capacity for spike days (Nov 25, Dec 2) by hiring seasonal fulfillment personnel in October. - Establish pre-season cash flow projections accounting for working capital needs during peak periods.

Expected Impact: Avoid stockouts during peak season (cost: lost sales + customer dissatisfaction). Improve inventory turnover during low seasons (freeing £500k–£1M in working capital).

3. Product Portfolio Optimization (Medium Priority)

Problem: 374 products (7% of catalog) have ≤2 orders in 25 months; slow-movers consume warehouse space and capital.

Actions: - Conduct detailed profitability analysis on slow-movers (StockCodes with ≤2 orders). Calculate contribution margin, storage cost, and handling cost. - Discontinue bottom 10% of SKUs with negative or near-zero contribution margin. Repurpose warehouse space. - Bundle slow-moving items with top sellers (e.g., Regency Cakestand 3 Tier) to increase attachment rates.

Expected Impact: Free 5–10% of warehouse space. Reduce inventory carrying cost by £50k–£100k annually.

4. Geographic Market Development (Low Priority)

Problem: UK accounts for 83% of revenue; high-growth markets (Australia +336%, Japan +123%) remain underpenetrated.

Actions: - Pilot localized marketing campaigns in Australia, Japan, and Benelux (Belgium, Netherlands) where growth rates exceed 40% YoY. - Invest £30k–£50k in SEO, local marketplace presence, and trade shows in fast-growth markets. - Establish regional pricing and language support to reduce friction in high-growth territories.

Expected Impact: Achieve 20% growth in international revenue (from £3.0M to £3.6M) within 12 months, reducing UK concentration risk.

5. Pricing Strategy Review (Low Priority)

Problem: Business relies on low-price-point items (<£5, 78.5% of revenue). Price-purchase frequency correlation is near-zero (-0.032), indicating minimal price sensitivity.

Actions: - Develop premium product tier (£15–£50 band) targeting institutional and bulk purchasers. Market as "wholesale partner solutions." - Test 5–10% price increases on top-selling, high-repeat products (Postage, White Hanging Heart T-Light Holder) where elasticity is low. - Create tiered discounts for volume purchases (£2k–£5k, £5k–£10k, >£10k) to drive larger basket sizes without margin erosion.

Expected Impact: Increase average order value from £470 to £520 (+11%), boosting revenue by £1.9M annually with minimal volume loss.

6. Data Analytics Infrastructure (Foundation)

Actions: - Implement automated monthly KPI dashboards tracking: customer segments, monthly revenue trends, product performance, geographic breakdowns, and inventory age. - Set up real-time alerts for anomalies: customer dormancy, product obsolescence, order volume spikes, and channel-specific underperformance. - Establish weekly reporting cadence for operations/sales teams, monthly steering committees for senior leadership.

Expected Impact: Reduce decision-making cycle time from quarterly to weekly. Enable proactive, data-driven management vs. reactive firefighting.

Conclusion

This analysis of a 2-year wholesale distribution operation reveals a business with clear operational strengths—consistent revenue, efficient product focus, strong seasonal patterns—alongside material risks and growth opportunities.

The core finding is customer concentration: Top 20% of customers drive 77% of revenue, and the Champions segment alone accounts for 72% of total value. This creates acute retention dependency. A systematic customer retention program targeting the Champions and At Risk segments is not optional—it is essential risk mitigation worth £1.7M+ in annual revenue protection.

Secondary opportunities exist in underutilized segments: 51% of the customer base is dormant, 374 SKUs have barely sold, and international markets are demonstrating triple-digit growth rates in emerging regions (Australia, Japan). Deploying capital toward product rationalization, geographic expansion, and pricing optimization can unlock £3M–£5M in incremental annual revenue without material operational disruption.

The data tells a consistent story: Low-price-point items dominate volume, high-repeat products drive loyalty, and Q4 seasonality is predictable and material. With this visibility, operational planning can shift from reactive to proactive. Inventory can be positioned in advance. Staffing can scale seasonally. Customer programs can be targeted by segment and risk profile.

Recommended next steps:

1. **Immediate (next 30 days):** Launch Champions account management program and dormancy detection system.
2. **Q1 planning:** Develop seasonal inventory and cash flow forecasts for next fiscal year.
3. **Q2 execution:** Begin slow-product discontinuation and geographic marketing pilots.
4. **Ongoing:** Establish weekly KPI dashboards and monthly steering reviews.

With disciplined execution of these recommendations, the organization can protect existing revenue streams, unlock growth in emerging markets, and transition from a data-informed to a truly data-driven operation.